

# Vol 11 Chapter 6 — K3 Surfaces, Niemeier Lattices, and the Leech Lattice

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

## Chapter 6 — K3 Surfaces, Niemeier Lattices, and the Leech Lattice

*Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine\_DIV5\_QM\_GR\_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.*

### Level 1 — one sentence

K3 surfaces (Kummer 1864, Kähler 1933, Kodaira 1964; L1 ESTABLISHED BST source) are compact complex surfaces with Hodge numbers  $h^{0,0} = h^{2,2} = 1, h^{2,0} = h^{0,2} = 1, h^{1,1} = 20$ , with K3 the load-bearing Bridge Object in BST (K57 RATIFIED, 7 L1 connections) reading quark masses via K3 Hodge — and the Leech lattice  $\Lambda_{24}$  (no-root Niemeier) is the  $\chi=24$  BST Bridge Object family member (K76).

### Level 2 — graduate-physicist precision

#### 6.1 K3 surfaces

A K3 surface is a compact complex surface with: - Trivial canonical bundle:  $K_X = \omega_X = 0$  -  $h^{0,1}(X) = h^{1,0}(X) = 0$  - Holomorphic 2-form  $\omega \in H^{0,2}$  exists (unique up to scale) - Hodge numbers:  $h^{0,0} = h^{2,2} = 1, h^{2,0} = h^{0,2} = 1, h^{1,1} = 20$

Euler characteristic:  $\chi(K_3) = 24$ . Signature:  $\sigma = 16$ .

All K3 surfaces are diffeomorphic but not all biholomorphic (60-real-dim moduli space).

Examples: quartic in  $\mathbb{P}^3$ , Kummer surface, elliptic K3, etc.

## 6.2 K3 in BST as load-bearing Bridge Object

K57 audit (RATIFIED, Spring 2026): K3 surface is one of three RATIFIED Bridge Objects in BST, with 7 L1 ESTABLISHED connections to other parts of the framework.

Tasks completed: - Task #91: Donaldson invariants of K3 via T1921 - Task #97: Mirror symmetry — K3 self-mirror map via BST - Task #119: K3 Hodge for quark cascade ( $m_c/m_s, m_b/m_c, m_t/m_b$ )

K3 is structurally load-bearing: removing it would collapse multiple BST predictions simultaneously.

## 6.3 Niemeier lattices

The 24 Niemeier lattices are the 24-dimensional even unimodular lattices (Niemeier 1973). Classified by their root system.

23 of them have root systems (A, D, E type). The 24th has no roots — the Leech lattice  $\Lambda_{24}$ .

## 6.4 Leech lattice $\Lambda_{24}$

Leech lattice (Leech 1967): unique even unimodular 24-dimensional lattice with no vectors of squared length 2.

Properties: - Densest sphere packing in 24D (Cohn et al. 2017 proof) - Kissing number:  $196560 = 2 \cdot 196883/2$  (related to Monster representation) - Automorphism group  $\text{Co}_0$  has order  $\sim 8 \times 10^{18}$  - Contains Mathieu group  $M_{24}$  as subgroup

## 6.5 Leech as BST Bridge Object (K76)

K76 audit-partial-ready (Spring 2026): Leech lattice  $\Lambda_{24}$  as BST  $\chi=24$  family Bridge Object member.

The 24 = Euler characteristic of K3; 24 = dimension of Leech; both anchor BST's  $\chi=24$  family of Bridge Objects.

Other  $\chi=24$  family members: 24-cell (K81),  $\Delta(\tau)$  modular discriminant (K82) — three independent  $\chi=24$  anchors (Cal #63 F1-F4 audit).

## 6.6 Mathieu groups and Niemeier

The 5 sporadic Mathieu groups  $M_{11}, M_{12}, M_{22}, M_{23}, M_{24}$  (L1 ESTABLISHED 1861-1873): smallest sporadic groups.

$M_{24} \subset$  Leech automorphism group  $\rightarrow \subset$  Monster (via Conway groups). Forms part of BST's L1  $\rightarrow$  L1.5  $\rightarrow$  convergence-hub architecture (Vol 11 Ch 12).

### 6.7 Mock modular forms (recent)

Mock modular forms (Zwegers 2002, Bringmann-Ono 2006): a holomorphic part of a real-analytic modular form. Connects to Ramanujan's mock theta functions.

Eichler-Zagier framework: mock modular  $\leftrightarrow$  Jacobi forms  $\leftrightarrow$  certain K3 sigma models.

BST May 2026 special-functions investigation includes mock theta as candidate substrate-natural special function (Vol 10 Ch 9).

### 6.8 K-audit anchors

- K3 Hodge 1962/64 (L1 ESTABLISHED)
- K57 (Spring 2026 RATIFIED): K3 as Bridge Object with 7 L1 connections
- K76 audit-partial-ready: Leech  $\Lambda_{24}$   $\chi=24$  family
- Tasks #91, #97, #119: K3 BST applications

### Level 3 — 5th-grader accessibility

K3 surfaces are special complex 4-real-dim manifolds with trivial canonical bundle. Discovered 1864 (Kummer), characterized 1962-64 (Kodaira). Hodge numbers  $(1, 20, 1)$  in  $H^{1,1}$  pattern; Euler char = 24. In BST: K3 is a Bridge Object (K57 RATIFIED) connecting 7+ different parts of the framework — load-bearing structural anchor. Reads quark masses via K3 Hodge (Task #119). Niemeier lattices: 24 even unimodular 24D lattices. The unique no-root one is the Leech lattice  $\Lambda_{24}$  — densest 24D sphere packing, contains Mathieu group  $M_{24}$ . Leech is the BST  $\chi=24$  Bridge Object family member (K76).

---

### What comes next

Chapter 7 develops Heegner numbers and Cremona elliptic curves.

### Where to look this up

- K3: Barth-Hulek-Peters-Van de Ven, *Compact Complex Surfaces*
- Niemeier/Leech: Conway-Sloane, *Sphere Packings, Lattices and Groups*
- BST: K57, K76, Tasks #91, #97, #119